Azal Soft Platform — Technology, verification and unit economics

TRIAD 2 · Tonight's case · answers prepared against the board template

Tonight's case · Technology, verification and unit economics

OUR ARCHITECTURE IN ONE DIAGRAM

Inputs → core → outputs. Draw it with shapes and arrows, or paste a photo of a sketch.

Meter photo at the well (agent's phone) → digit-checked reading (6 or 7 digits, operator confirms) → cycle ledger in Firestore, one chain per well → owner dashboard and CSV/PDF export.
No smart meter, no SIM, no extra energy: the phone camera is the sensor.

WHAT IS OURS

Proprietary: what we built and others cannot copy easily.

The cycle and loss engine: reading capture, consumption, rollover (closing = next opening), loss per cycle, subscriber and well ledgers.

WHAT WE PARTNER FOR OR BUY

Components, data or services we rely on others for.

Firebase (Auth, Firestore, rules, hosting), a vision model for the digits, and the agent's Android phone.

WHY IT IS DEFENSIBLE

Patent, data, know-how, certification, a partnership, or a cost no one else can match.

Not the camera — the rule set. An unbroken reading chain per well plus a loss figure per cycle produces a record that cannot be reconstructed afterwards, so every well we add deepens the baseline and the switching cost. Cost is also defensible: a phone and a photo instead of a smart meter and a SIM.

OUR VERIFICATION PLAN

What we will test, who checks it, by when, and the standard or certification we aim for.

1 · Accuracy. A labelled set of real meter photos, scored per digit count (6 vs 7) and per lighting condition — the accuracy number is published, not claimed.
2 · Integrity. The server rejects any cycle that opens at a reading other than the previous one's closing; every reading keeps operator and timestamp.
3 · Field proof. A parallel run at one real well for one full cycle, alongside the paper register, witnessed by a local partner (water department or university).
4 · Third party. Same evidence pack handed to an independent auditor once the parallel run is clean — the cheap order of de-risking.

UNIT ECONOMICS · one unit, what it costs us, what we charge

OUR UNIT	COST PER UNIT	PRICE PER UNIT	MARGIN PER UNIT
Per m³, per site, per connection.	*All it costs us to deliver one.*	*What the customer pays.*	*Price minus cost.*
One well · one month (≈20 subs, ≈20 cycles)	**$1.20** OCR + hosting + support	**$12** per well / month	**90%** ≈$10.8 per well / month

FOR YOUR LANE — DIGITAL, NOT HARDWARE

Hardware: energy per m³ (kWh). Digital: the cost to serve one more customer.

The cost to serve one more well is ≈$1 a month: no hardware, no energy per m³, no installation. That is the whole point of the model — the marginal cost is flat while the recovered loss scales with every well. Pricing follows the customer's avoided loss, not our server bill.

Two numbers to keep honest in front of the jury: **accuracy** (measured on a labelled photo set) and **cost per unit** (what it costs us, not what the customer pays). Replace every example figure above with your own before the session.